PRODUCT USER MANUAL

OMI_TEMPSAL
Sea Surface Temperature
BLKSEA_OMI_TEMPSAL_sst_area_averaged_anomalies
BLKSEA_OMI_TEMPSAL_sst_trend
Issue 5.0

Contributors: **Andrea Pisano, Claudia Fanelli, Claudia Cesarini, Flavio La Padula, Bruno Buongiorno Nardelli**

Approval Date: November 2024

MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

Record Table

Issue	Date	§	Description of Change	Author	Checked by	Validated by
1.0	2018/10/09	All	Creation of the document	Andrea Pisano, Bruno Buongiorno Nardelli		
2.0	2019/02/20	All	Reviewed document	Andrea Pisano, Bruno Buongiorno Nardelli		
3.0	2020/04/14	All	Updated time coverage/filenames	Andrea Pisano, Bruno Buongiorno Nardelli		
3.1	2020/09/10	All	Update of the upstream reference dataset and time extension to 2019	Andrea Pisano, Bruno Buongiorno Nardelli		
3.2	2021/12/14	All	Time extension to 2020	Andrea Pisano, Claudia Fanelli, Bruno Buongiorno Nardelli		
4.0	2022/06/16	All	Addition of the year 2021 + automatic temporal extension	Andrea Pisano, Claudia Fanelli, Claudia Cesarini, Flavio La Padula, Bruno Buongiorno Nardelli	A. Pisano	
5.0	2024/05/31	All	Update for 2023; Change of climatology to 1991-2020; Backward extension to 1982	Andrea Pisano, Claudia Fanelli, Claudia Cesarini, Flavio La Padula, Bruno Buongiorno Nardelli	A. Pisano	

GLOSSARY AND ABBREVIATIONS

CMEMS	Copernicus Marine Environment Monitoring Service
MO	Mercator Ocean
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
SST	Sea Surface Temperature
BLK	Black Sea
REP	Reprocessed

Table of Content

GLOSSARY AND ABBREVIATIONS **3**

I INTRODUCTION **6**

I.1. Summary **6**

I.2. History of changes **7**

II DESCRIPTION OF THE PRODUCT **8**

II.1. General information **8**

II.2. Details of the datasets **9**

II.2.1. BLKSEA_OMI_TEMPSAL_sst_area_averaged_anomalies 9

II.2.2. BLKSEA_OMI_TEMPSAL _sst_trend 9

II.3. Product System Description **9**

II.3.1. BLKSEA_OMI_TEMPSAL_sst_area_averaged_anomalies 9

II.3.2. BLKSEA_OMI_TEMPSAL_sst_trend 10

III FILE FORMAT **10**

IV FILES NOMENCLATURE AND FORMAT **10**

IV.1. Nomenclature of files **10**

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1. Summary

This document is the user manual for the CMEMS Sea Surface Temperature (SST) Ocean Monitoring Indicator (OMI) for the Black Sea (BLKSEA). The BLKSEA SST OMI belongs to the ocean temperature and salinity (TEMPSAL) OMI family and provides a 1D time series of area averaged monthly mean SST anomalies and a 2D map of SST trend for the Black Sea. SST anomalies are computed with respect to the 1991-2020 reference period (climatology). This climatology is selected according to WMO recommendations (WMO, 2017) and the most recent practice from the U.S. National Oceanic and Atmospheric Administration practice (https://wmo.int/media/news/updated-30-year-reference-period-reflects-changing-climate). The BLKSEA SST OMI is built from the CMEMS Reprocessed Black Sea SST L4 (BS-REP-L4) product[1] (see corresponding PUM[2] and QUID[3]) and its update runs in parallel with the corresponding BS-REP-L4 update.
Specifically, the BLKSEA SST OMI includes two products/datasets (here, product name coincides with dataset name):

- **BLKSEA_OMI_TEMPSAL_sst_area_averaged_anomalies**, which provides unfiltered and filtered monthly mean anomaly time series of SST averaged over the Black Sea;
- **BLKSEA_OMI_TEMPSAL_sst_trend**, which contains the 2D map of annual cumulative/net trend over the Black Sea.

The BLKSEA SST OMI products are published on the CMEMS dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.4 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The BLKSEA SST OMI products are also described in the Quality Information Document (QUID) CMEMS-OMI-QUID-BLKSEA-SST.pdf (http://marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-BLKSEA-SSTpdf). See also News flash. More detailed information can be obtained from http://marine.copernicus.eu/services-portfolio/contact-us/

[1] cmems_SST_BS_SST_L4_REP_OBSERVATIONS_010_022
[2] http://marine.copernicus.eu/documents/PUM/CMEMS-SST-PUM-010-021-022-041-042.pdf
[3] http://marine.copernicus.eu/documents/QUID/CMEMS-SST-QUID-010-021-022-041-042.pdf

I.2. History of changes

2018/09	Publication of BLKSEA SST OMI (BLKSEA_OMI_TEMPSAL_sst_area_averaged_anomalies; BLKSEA_OMI_TEMPSAL_sst_trend)
2020/04	Addition of the year 2018
2020/09	Update of the upstream reference dataset and addition of the year 2019
2021/12	Addition of the year 2020
2022/06	Addition of the year 2021 + automatic temporal extension
2024/09	Temporal extension (up to 2023), Backward temporal extension to 1982, New climatology 1991-2020 (30 years)

II DESCRIPTION OF THE PRODUCT

II.1. General information

Product(s) name	BLKSEA_OMI_TEMPSAL_sst_area_averaged_anomalies
Geographical coverage	Black Sea (38° N 43' – 48°N 46'; 26°E 22' – 42°E 22')
Variables	Unfiltered and filtered time series of SST anomalies sst_anomaly [°C] sst_anomaly_filtered [°C]
Update frequency	Annual
Available time series	1982-01-01 to year-1
Delivery mechanism	MDS
Temporal resolution	Monthly averages
Spatial resolution	Not applicable
Integration depth (for example)	Surface
Format	NetCDF CF1.7

Product(s) name	BLKSEA_OMI_TEMPSAL_sst_trend
Geographical coverage	Black Sea (38° N 43' – 48°N 46'; 26°E 22' – 42°E 22')
Variables	Map of SST cumulative trend sst_trend [°C]
Update frequency	Annual
Available time series	1982-01-01 to year-1
Delivery mechanism	MDS
Temporal resolution	Time integrated
Spatial resolution	0.05° x 0.05°
Integration depth (for example)	Surface
Format	NetCDF CF1.7

II.2. Details of the datasets

For each product, one table with details of datasets content.

II.2.1. BLKSEA_OMI_TEMPSAL_sst_area_averaged_anomalies

BLKSEA_OMI_TEMPSAL_sst_area_averaged_anomalies
Contains the 1D monthly mean SST anomaly time series (unfiltered and filtered) averaged over the Black Sea and derived from the CMEMS BS-REP-L4 SST product (cmems_SST_BS_SST_L4_REP_OBSERVATIONS_010_022). Anomalies are computed with respect to the 1991-2020 reference period. The filtered anomaly is obtained by using the X11-seasonal adjustment procedure.
sst_anomaly [°C] long_name = "area averaged sea surface temperature anomaly" standard_name = "sea_surface_temperature"
sst_anomaly_filtered [°C] long_name = "area averaged sea surface temperature filtered anomaly" standard_name = "sea_surface_temperature"

II.2.2. BLKSEA_OMI_TEMPSAL _sst_trend

BLKSEA_OMI_TEMPSAL_sst_trend
Contains the 2D map of annual cumulative/net SST trend over the Black Sea and derived from the CMEMS BS-REP-L4 SST product (cmems_SST_BS_SST_L4_REP_OBSERVATIONS_010_022).
sst_trend [°C] long_name = "cumulative sea surface temperature trend" standard_name = "change_over_time_in_sea_surface_temperature"

II.3. Product System Description

II.3.1. BLKSEA_OMI_TEMPSAL_sst_area_averaged_anomalies

This OMI product includes unfiltered SST anomalies, given as monthly mean time series starting on 1982 and averaged over the Black Sea, and filtered SST anomalies, obtained by using the X11-seasonal adjustment procedure. This OMI is derived from the CMEMS BS-REP-L4 SST product (cmems_SST_BS_SST_L4_REP_OBSERVATIONS_010_022), which provides the SSTs that are used to compute the evolution of SST anomalies (unfiltered and filtered) over the Black Sea. Anomalies are computed against the 1991-2020 reference period. The 30-year climatology 1991-2020 is defined according to the WMO recommendation (WMO, 2017) and recent U.S. National Oceanic and

Atmospheric Administration practice (https://wmo.int/media/news/updated-30-year-reference-period-reflects-changing-climate).

II.3.2. BLKSEA_OMI_TEMPSAL_sst_trend

This OMI product includes the cumulative/net trend for the Black Sea, i.e. the rate of change (°C/year) scaled by the number of time steps (namely, the number of years). Trend computation is based on the X-11 seasonal adjustment procedure. This OMI is derived from the CMEMS BS-REP-L4 SST product (cmems_SST_BS_SST_L4_REP_OBSERVATIONS_010_022).

III FILE FORMAT

The products are stored using the NetCDF format. To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE AND FORMAT

Information about nomenclature of files when downloaded can be found in this articles

" How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

IV.1. Nomenclature of files

The BLKSEA SST OMI files nomenclature when downloaded through the CMEMS MDS Services is based on the following nomenclature:

<datasetname>_<timestep>_R<referenceclimatology>_P<productiontime>.nc

Where :

- **<datasetname>** is the dataset reference.
- **<timestep>** is the first time step contained in the file.

- **<referenceclimatology>** is the climatology with which the anomalies are computed.
- **<productiontime>** is the production time when the file was generated.
- **.nc** is the standard NetCDF filename extension.

Examples of the file names are:

- blksea_omi_tempsal_sst_area_averaged_anomalies_19820101-20231231_R19912020_P20240501.nc
- blksea_omi_tempsal_sst_trend_2023_R19912020_P20240501.nc